

Statistical Modelling and Inference for Health – Assessment 1

Student ID: 14210037

Introduction

Rheumatoid arthritis (RA) is a systemic autoimmune disease characterised by symmetrical joint pain and morning stiffness. Non-steroidal anti-inflammatories and corticosteroids are prescribed to alleviate pain, while the early initiation of disease-modifying antirheumatic drugs can prevent long-term joint damage. There is a strong interest in a relationship between the initial diagnostic assessment using the Disease Activity Score-28 (DAS28) and subsequent Visual Analogue Scale (VAS) scores. Therefore, this study aims to examine the association between baseline DAS28 and pain progression over time.

Methods

This longitudinal cohort study included 500 patients with RA, each followed for up to 10 visits. At baseline, age, DAS28, and VAS were recorded; only VAS was measured on subsequent visits. Exploratory analyses summarised baseline age and DAS28 using histograms, and visualised pain trajectories with a line plot of VAS over visit number, coloured by patient ID, alongside box plots showing the mean and quartiles. Pearson correlation coefficients were calculated to examine the relationship between key variables.

VAS trajectories were then modelled using linear mixed-effects models with patients as the clustering factor. Random-intercept models were first fitted and then extended to random-slope models to allow patient-specific slopes. To address the clinical question, an interaction between baseline DAS28 and visit number was included to assess whether baseline DAS28 modified the rate of change in VAS over time. Then, baseline age was then introduced as a potential confounder, and models with and without age were compared. Model assumptions were assessed using residual-versus-fitted plots, QQ-plots of residuals and random effects, and residual plots against visit number and baseline DAS28.

Results

1. Population Characteristic

There was no missing data across the ten visits for any participant. Table 1 illustrates baseline characteristics of the 500 participants. The average age was 60.6 years (SD = 10.2), with a median of 60.9 years, and a wide age range spanning from 32.3 years in adulthood to 93.7 years in elderly. Mean DAS28 was 4.09 (SD = 1.19), with values ranging from 1.35 to 8.62 and a median of 3.94.

Table 1. Baseline characteristics of participants (N = 500)

Variable	Mean (SD)	Median (IQR)	Range
Age (years)	60.6 (10.2)	60.9 (53.1, 67.2)	32.3 - 93.7
DAS28	4.09 (1.19)	3.94 (3.24, 4.89)	1.35 - 8.62

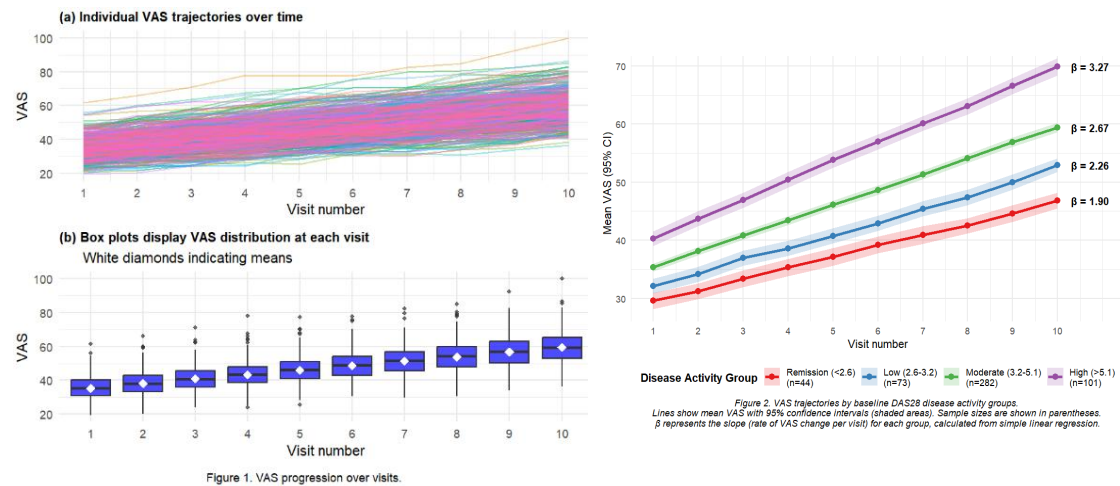
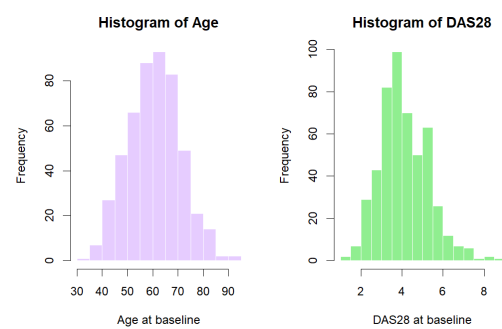
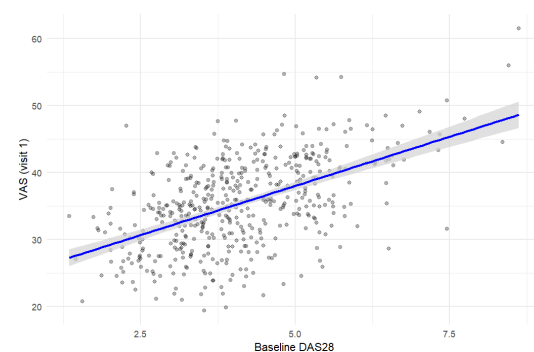


Figure 1 presents the change in VAS scores across visit times, with both the line plot and box plots indicating a gradual increase over time. Figure 2 shows the pain scores over visit time, stratified by disease activity groups defined by baseline DAS28. Higher baseline disease activity is associated with a steeper increase in VAS, while the sample is concentrated in the moderate ($n = 282$) and high ($n = 101$) disease activity groups.



through time as outcome, both were treated as continuous variables in the linear mixed model.



The weak correlation between baseline age and DAS28 ($r = -0.053$, 95% CI -0.140 – 0.035 , $p = 0.238$) indicates minimal collinearity between these predictors. Age showed a weak positive correlation with baseline VAS ($r = 0.519$, 95% CI 0.451 – 0.580 , $p < 0.001$), suggesting that older patients tended to report slightly higher pain levels at treatment initiation. Baseline DAS28 and baseline VAS demonstrated a moderate positive correlation ($r = 0.524$, 95% CI 0.457 – 0.585 , $p < 0.001$), providing preliminary evidence of an association between disease activity and pain intensity. Figure 3 presents the relationship between baseline DAS28 and VAS. Blue line fits suggest an approximately linear positive association, supporting the use of linear mixed models for subsequent analysis.

2. Model Selection

Table 2. Comparison of linear mixed-effects models

Model	Fixed Effects	Random Effects	Log-Likelihood	AIC	BIC
Random intercept only	Age + DAS28 + Visit	(1 ID)	-12932.6	25877.2	25916.3
Random intercept + slope	Age + DAS28 + Visit	(1 + Visit ID)	-12337.9	24691.8	24743.9
Random intercept + slope + interaction	Age + DAS28 × Visit	(1 + Visit ID)	-12158.3	24334.6	24393.2

Note:

Lower AIC/BIC values indicate better model fit.

Model comparison showed that with patient-specific slope significantly improved fit compared to the random intercept-only model. This suggests considerable inter-individual heterogeneity in disease progression. The selected model, including random intercept, random slope for visit number, and DAS28 with Visit interaction, showed the best fit (lowest AIC=24334.6, and BIC=24393.2, with highest log-likelihood = -12158.3), substantially lower than simpler alternatives.

Table 3. Parameter estimates from the final linear mixed-effects model

Parameter	Estimate	95% CI
Fixed Effects		
Intercept	0.05	(-2.40, 2.50)
Age (per year)	0.35	(0.32, 0.38)
DAS28 (per unit)	2.80	(2.47, 3.12)
Visit number (per visit)	1.15	(1.02, 1.28)
DAS28 × Visit number	0.37	(0.34, 0.40)
Random Effects		
Between-patient SD (intercept)	4.09	-
Between-patient SD (slope for visit)	0.35	-
Correlation (intercept-slope)	-0.41	-
Residual SD	2.19	-

Table 3 shows parameter estimates from the final model. The intercept (0.05, 95% CI: -2.40–2.50) was non-significant, as expected for this theoretical reference condition. The coefficient for age was approximately 0.35 (95% CI 0.32–0.38), meaning that, for patients with the same baseline DAS28 and visit number, each additional year of age was associated with an increase of about 0.35 units in mean VAS, indicating a positive association between age and pain. The main effect of DAS28 was larger: each one-unit higher baseline DAS28 was associated with roughly 2.5–3 additional VAS units, suggesting that

patients with higher disease activity at baseline experienced substantially more pain. On average, VAS increased by 1.15 units per visit. However, the significant interaction between DAS28 and visit number was 0.37 (95% CI 0.34–0.40), indicating that patients with higher baseline DAS28 experienced faster pain progression over time. For example, holding other parameters constant, when DAS28 = 2, the expected increase in VAS per visit is about 1.89 (= 1.15 + 2 × 0.37). While DAS28 = 6, it rises to about 3.37 units per visit (= 1.15 + 6 × 0.37), showing that higher DAS28 is associated with rapid worsening of pain.

The random-effects estimates indicated substantial between-patients variability in baseline VAS (SD = 4.9) and in the individual time slopes (SD = 0.35 units per visit), while the

negative correlation between the random intercept and slope suggested that patients starting with higher baseline VAS tended to have slightly flatter trajectories over time.

To assess whether age confounded the association between baseline DAS28 and VAS, we compared two random-slope models: one including DAS28 and visit number only, with another additionally adjusting for baseline age. In the model without age, each one-unit increase in baseline DAS28 was associated with an average increase of 4.664 units in VAS, whereas after adjustment for age, the corresponding estimate was 4.652, a relative change of 0.25%. By contrast, a likelihood ratio test comparing these models showed that adding age substantially improved model fit ($\Delta AIC = 316$, $p < 0.001$), indicating that age is an important independent predictor of VAS but does not materially confound the DAS28–VAS association.

The final formula:

$$VAS_{ij} = \beta_0 + 0.35 Age_i + 2.80 DAS28_i + 0.37 \times (DAS28_i \times Visit_{ij}) + (1.15 + b_{1i}) \times Visit_{ij} + b_{0i} + \varepsilon_{ij}$$

$b_{0i} \sim N(\theta, \sigma_{b_0}^2)$ represents patient – specific random intercept

$b_{1i} \sim N(\theta, \sigma_{b_1}^2)$ represents patient – specific random slope for visit

$\varepsilon_{ij} \sim N(\theta, \sigma^2)$ represents residual error

3. Model Diagnostics

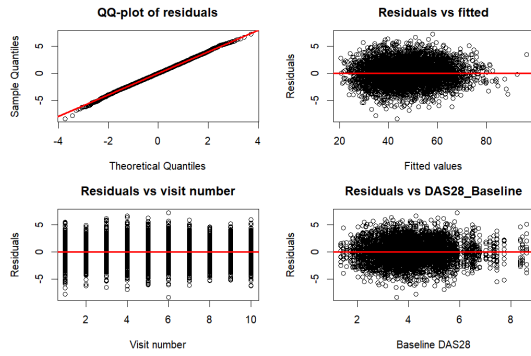


Figure 5. Residual diagnostics

Figure 5 presents residual diagnostics for the final model. The QQ-plot of residuals lies close to the 45-degree reference line, with only mild deviation in the extreme tails, supporting the assumption of approximately normally distributed residuals. The residuals-versus-fitted plot shows a roughly symmetric cloud around zero with no clear pattern or funnel shape. Residuals plot against visit number and baseline DAS28 showed no systematic patterns, supporting the assumptions of linear time trends and linear DAS28 effects, respectively. The constant spread of residuals across both predictors confirmed homoscedasticity.

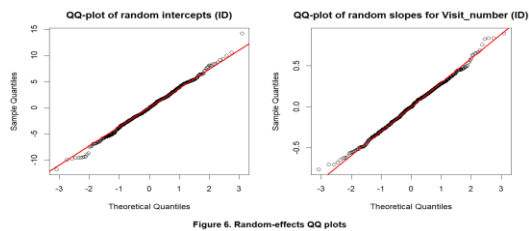


Figure 6. Random-effects QQ plots

supporting the normality assumption for random effects.

Figure 6 shows QQ-plots of the model random intercepts and random slopes for visit number. In both cases, most points lie near the theoretical normal line, with only a few extreme observations deviating in the tails, supporting the normality assumption for random effects.

Conclusion

In conclusion, this study explored how pain that individual felt was related to baseline DAS28, age and follow-up time in a longitudinal cohort of patients with RA. The linear mixed-effects model quantified how strongly each of these factors influenced VAS, and showed that baseline DAS28 not only affected baseline pain levels but also modified the rate of pain worsening over time, with higher DAS28 values associated with steeper increases in VAS across visits (interaction coefficient = 0.37, 95% CI 0.34-0.40). Moreover, the random-effects structure revealed a negative correlation (-0.41) between patient-specific intercepts and slopes, suggesting that patients starting with higher VAS tended to have flatter trajectories, possibly reflecting ceiling effects or differential treatment responses.

Discussion

Clinically, these findings highlight that patients with high disease activity at treatment initiation are at greater risk of persistent and worsening pain. In practice, patients with higher baseline DAS28 should be considered for intensive monitoring and early treatment escalation, supported by this research finding.

Several limitations of the data should be acknowledged. First, important clinical characteristics were not available, including gender (RA is more prevalent in women), comorbidities, medication history, and family history, any of which could influence both DAS28 and pain or as unmeasured confounders. Second, VAS was treated as a continuous outcome with approximately normal residuals. Although diagnostics supported this assumption, some extreme observations suggest that future studies with larger sample sizes might benefit from other complex mixed-effects models. From an analytical perspective, only relatively simple random-effects structures and linear time trends were evaluated. Future research could explore detailed time patterns, or more complex random-effects structures that allow between-patient variation in the DAS28 effect. The correlation structure between random effects was assumed to be constant across patients and that residual variance was homogeneous over time. More flexible covariance structures could be explored in future work.

Despite these limitations, the analysis demonstrates how mixed-effects modelling can be used to answer a clinically relevant question about the relationship between disease activity and pain progression, and provides a framework that could be applied to richer clinical datasets in future work.